

CO2-AT1-SCENARIO BASED PROBLEMS

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COURSE CODE: CSA0911

QUESTION-1:

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

1/5 answered

Q1 Student Registration System10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class StudentRegistration {
3     public static void main(String[] args) {
4         HashSet<Student> students = new HashSet<>();
5         Student s1 = new Student(101, "Rakshana");
6         Student s2 = new Student(102, "Priya");
7         students.add(s1);
8         students.add(s2);
9         HashMap<Student, List<String>> courses = new HashMap<>();
10        courses.put(s1, Arrays.asList("Java", "DBMS"));
11        courses.put(s2, Arrays.asList("Python", "AI"));
12        Iterator<Student> itr = students.iterator();
13        while (itr.hasNext()) {
14            Student s = itr.next();
```

Input

stdin input

Output

```
Student : 101 Rakshana
Courses : [Java, DBMS]

Student : 102 Priya
Courses : [Python, AI]
```

QUESTION-2:

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q2 E-Commerce Cart System10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2 public class ShoppingCart {
3     public static void main(String[] args) {
4         ArrayList<String> cart = new ArrayList<>();
5         cart.add("Laptop");
6         cart.add("Mouse");
7         cart.add("Keyboard");
8         cart.add("Headset");
9         Iterator<String> itr = cart.iterator();
10        System.out.println("Items in Cart:");
11        while (itr.hasNext()) {
12            System.out.println(itr.next());
13        }
14    }
```

Input

stdin input

Output

```
Items in Cart:
Laptop
Mouse
Keyboard
Headset
```

QUESTION-3:

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.*;
2
3 public class LibraryManagement {
4
5     public static void main(String[] args) {
6
7         HashMap<String,String> books = new HashMap<>();
8
9         books.put("97812345","Java Programming");
10        books.put("97867890","Python Basics");
11        books.put("97854321","Data Structures");
12
13        System.out.println("Retrieved Book:");
14        System.out.println(books.get("97867890"));
```

Input

stdin input

Output

```
Retrieved Book:
Python Basics

All Books:
97854321 -> Data Structures
97812345 -> Java Programming
```

QUESTION-4:

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2 public class EmployeeProcessing{
3     public static void main(String[] args){
4         ArrayList<Employee> list=new ArrayList<>();
5         list.add(new Employee(101,"Ravi",45000));
6         list.add(new Employee(102,"Priya",65000));
7         list.add(new Employee(103,"Rakshana",70000));
8         Iterator<Employee> itr=list.iterator();
9         System.out.println("Employees Salary > 50000");
10        while(itr.hasNext()){
11            Employee e=itr.next();
12            if(e.salary>50000){
13                System.out.println(e);
14            }
15        }
```

Input

stdin input

Output

```
Employees Salary > 50000
102 Priya 65000.0
103 Rakshana 70000.0
```

QUESTION-5:

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```
1 import java.util.*;
2 public class OnlineVoting {
3     public static void main(String[] args) {
4
5         HashSet<String> voters = new HashSet<>();
6
7         HashMap<String,Integer> votes = new HashMap<>();
8
9         if(voters.add("Voter1"))
10             votes.put("Candidate A",
11                 votes.getDefault("Candidate A",0)+1);
12
13         if(voters.add("Voter2"))
14             votes.put("Candidate B",
15                 votes.getDefault("Candidate B",0)+1);
```

Input

stdin input

Output

```
Duplicate Vote Not Allowed

Voting Results
Candidate B = 1
Candidate A = 2
```